

WANCONG ZHANG

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Education

PhD in Computer Science, New York University, Courant Institute *09/2023-Present*
Advisor: Prof. Yann LeCun

MS in Computer Science, New York University, Courant Institute *09/2018-09/2020*
GPA: 3.9 / 4.0

Research / Employment

Research Scientist Intern, Meta FAIR (Video-SSL Group) *05/2025-Present*
Advised by Dr. Nicolas Ballas

- Hierarchical world models for long horizon planning in video-based control tasks.

PhD Candidate, NYU *09/2023-Present*
Advised by Prof. Yann LeCun

- Joint embedding prediction architecture (JEPA) for zero-shot planning. [Paper](#)
- Designed a probing benchmark for unsupervised visual representations in RL, improving Atari pretraining performance beyond SOTA. [Paper](#)

Senior Deep Learning Researcher, AssemblyAI *02/2021-08/2023*

- Spearheaded the development of industry-leading automatic speech recognition (ASR) models ([Conformer-1](#), [Conformer-2](#), realtime ASR).
- Directed teams of up to 10 researchers/engineers, deploying multi-node training pipelines at scale (Google Cloud + on-prem clusters).

M.S. Researcher, NYU *09/2019-06/2020*
Advised by Profs Sam Bowman & Rajesh Ranganath

- Proposed MixUp algorithms for sentence classification using transformers, reducing calibration error by 50%. [preprint](#)
- Built customized, fused CUDA kernels for a novel gateless RNN architecture. [Blog](#)

Research Intern, University of Toronto *06/2019-09/2019*
Advised by Prof. Marzyeh Ghassemi

- Multimodal representation learning (labs + clinical notes) for healthcare. [Paper](#)

Software Engineer, Guardant Health *07/2017-07/2018*

- Fullstack web development and distributed systems

Peer Reviewed Publications

Learning from Reward-Free Offline Data: A Case for Planning with Latent Dynamics Models
W Zhang*, V Sobal*, K Cho, R Balestriero, T Rudner, Y LeCun
Advances in Neural Information Processing Systems (**NeurIPS**), 2025 [[Paper](#)]
Best Paper Award, ICML 2025 Workshop on Building Physically Plausible World Models

Light-weight probing of unsupervised representations for Reinforcement Learning
W Zhang, A Chen, V Sobal, Y LeCun, N Carson
Reinforcement Learning Conference (**RLC**) 2024 [[Paper](#)]

A Comprehensive EHR time series pre-training benchmark
M McDermott, B Nestor, **W Zhang**, P Szolovitz, A Goldenberg, M Ghassemi
The ACM Conference on Health, Inference, and Learning (**CHIL**) 2021 [[Paper](#)]

Preprints

Conformer-1: Robust ASR via Large-Scale Semisupervised Bootstrapping
W Zhang, L Chkhetiani, F Ramirez, Y Share, A Vanzo, M Liang, S Martin, G Oexle, R Bousbib, T
Pesach, M Nguyen, D Pulliam, D Donato
arXiv preprint 2024 [[arXiv](#)]

Mixup training leads to reduced overfitting and improved calibration for the transformer
architecture
W Zhang, I Vaidya
arXiv preprint 2021 [[arXiv](#)]

Teaching

Graduate Student Instructor: [Deep Learning](#) (NYU) *Fall 2024*

- Designed and hosted the final course competition for ~150 students ([project page](#))
- Designed homework assignments and coordinated graders
- Supervised M.S. students with independent research projects on video SSL